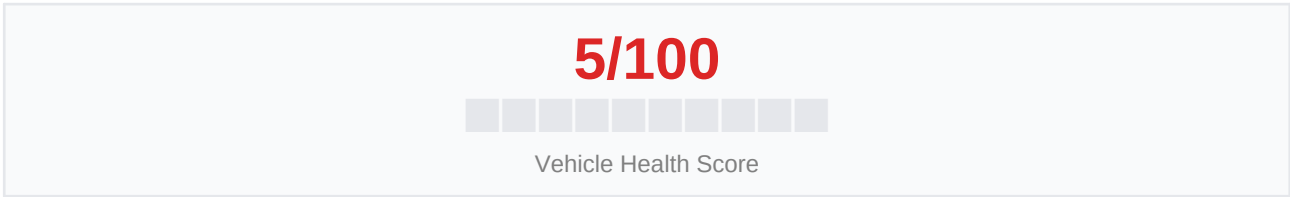


VEHICLE INSPECTION REPORT

AI-Powered Body + Engine Sound Analysis · June 11, 2026 at 21:54
Physical Defect Detection (7-class YOLO + Severity Model) + Engine Knock Audio Classification

VIN / Registration	JTHBK1GG7R1234567	Mileage	45000
Make / Model	LEXUS LEXUS-SUV-TEST	Year	2024

AI Vehicle Health Assessment



Risk Level	High	Overall Status	FAIL
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Key Risk Factors:

- Safety-critical components affected: Windshield — Glass Crack, Lamp Crack, Bonnet — Moderate Dent, Glass Crack
- High-confidence damage on: Windshield — Glass Crack, Bumper — Severe Dent, Lamp Crack, Bonnet — Moderate Dent, Glass Crack
- Minor cosmetic damage on: Fender — Moderate Dent, Moderate Dent

AI Inspection Summary

The inspection detected: Windshield — Glass Crack, Bumper — Severe Dent, Lamp Crack, Bonnet — Moderate Dent, Fender — Moderate Dent, Moderate Dent, Glass Crack. Safety-critical damage identified on Windshield — Glass Crack, Lamp Crack, Bonnet — Moderate Dent, Glass Crack requires immediate attention.

Defect Analysis — Dual Model Results

- Windshield — Glass Crack ■ Safety-Critical** (Severe, 99.9% confidence)
Windshield cracks impair visibility and cabin structural integrity — repair or replace immediately.
- Bumper — Severe Dent** (Severe, 96.5% confidence)
Bumper damage reduces collision absorption — repair recommended.
- Lamp Crack ■ Safety-Critical** (Moderate, 72.3% confidence)
Cracked lamp covers allow moisture ingress — replace promptly.
- Lamp Crack ■ Safety-Critical** (Minor, 52.6% confidence)
Cracked lamp covers allow moisture ingress — replace promptly.

Bonnet — Moderate Dent ■ Safety-Critical (Severe, 86.4% confidence)
Bonnet damage may indicate impact near the engine — inspect for underlying mechanical damage.

Bumper — Severe Dent (Moderate, 79.6% confidence)
Bumper damage reduces collision absorption — repair recommended.

Fender — Moderate Dent (Moderate, 66.0% confidence)
Exposed metal will oxidise — treat and repair at nearest opportunity.

Moderate Dent (Moderate, 57.5% confidence)
Moderate dent detected — repair recommended to prevent further damage.

Bumper — Severe Dent (Severe, 91.4% confidence)
Bumper damage reduces collision absorption — repair recommended.

Glass Crack ■ Safety-Critical (Severe, 86.2% confidence)
Glass cracks compromise structural integrity and visibility — replace immediately.

Lamp Crack ■ Safety-Critical (Severe, 85.9% confidence)
Cracked lamp covers allow moisture ingress — replace promptly.

AI Recommendations

- 1. Windshield cracks impair visibility and cabin structural integrity — repair or replace immediately.
- 2. Bumper damage reduces collision absorption — repair recommended.
- 3. Cracked lamp covers allow moisture ingress — replace promptly.
- 4. Bonnet damage may indicate impact near the engine — inspect for underlying mechanical damage.
- 5. Exposed metal will oxidise — treat and repair at nearest opportunity.

Engine Sound Analysis

Engine Status	ENGINE HEALTHY	Confidence	73.19%
Verdict	clean	Duration	11.81s
Remarks	No knock detected. Engine sounds healthy. Continue regular maintenance schedule.		

Standard 7-Part Damage Checklist

Component	Defect Type / Severity	Confidence	Status
Bonnet	Bonnet — Moderate Dent	86.4%	Severe ■
Bumper	Bumper — Severe Dent	96.5%	Severe
Dickey	No Damage Detected	—	—

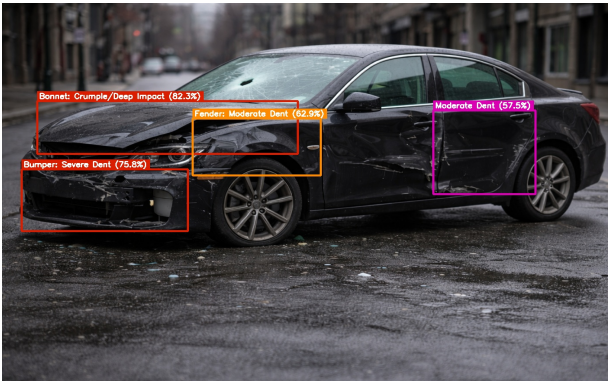
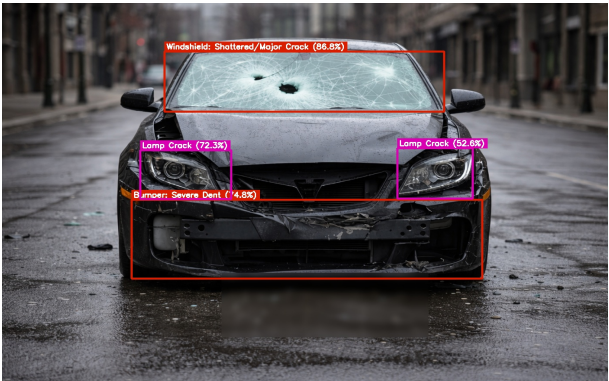
Door	No Damage Detected	—	—
Fender	Fender — Moderate Dent	66.0%	Moderate
Light	No Damage Detected	—	—
Windshield	Windshield — Glass Crack	99.9%	Severe ■
Total Unique Parts/Types	7	Overall Status:	FAIL

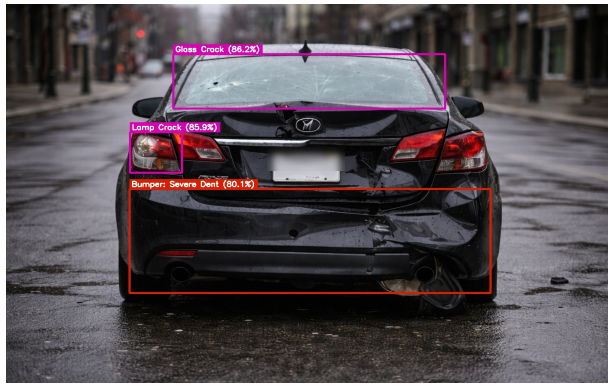
Additional Damage Detections (Severity Model)

Damage Type	Confidence	Severity Tier
Lamp Crack	72.3%	Moderate
Lamp Crack	52.6%	Minor
Moderate Dent	57.5%	Moderate
Glass Crack	86.2%	Severe
Lamp Crack	85.9%	Severe

Annotated Images — Dual Model Detections

Blue/green boxes = part detections; Purple boxes = severity-only detections





This report was generated using the unified MEHRA inspection pipeline: physical defect detection with a 7-class part model and severity model, engine sound knock classification, and Groq LLaMA 3.3 for report analysis. Results should be verified by a qualified technician.